

TD3 - Cassandra

5 Activities on querying and data models

5.1 Queries (in CQL)

1. All movies Requete :

```
SELECT * FROM movie;
```

Retour :

idimdb	budget	poster
tt5715410	30000000	https://image.tmdb.org/t/p/original/5yCQvhFuFPY0pACxoUpLULYCxop.jpg
tt12441478	0	https://image.tmdb.org/t/p/original/86jyzdxtAX8p956rVQDKwGbrYh2.jpg
tt0266543	94000000	https://image.tmdb.org/t/p/original/eHuGQ10FUzK1md0Y69wF5pGgEf5.jpg
tt0126029	60000000	https://image.tmdb.org/t/p/original/iB64vpL3dIObOtMZgX3RqdVdQDc.jpg
tt1453405	200000000	https://image.tmdb.org/t/p/original/upaMBEonGeZGQc3xtUVZ0fe1lG1.jpg
tt6718170	100000000	https://image.tmdb.org/t/p/original/qNBAXBIQln0ThrVvA6mA2B5ggV6.jpg
tt0413267	160000000	https://image.tmdb.org/t/p/original/n4SexGGQzI26E269tfpa80MZaGV.jpg
tt1754455	0	https://image.tmdb.org/t/p/original/ot2u06E1TjVCXkcoAEPSE1hHuTi.jpg
tt0448115	80000000	https://image.tmdb.org/t/p/original/xnopI5Xtky18MPPhK40cZAGA0VeV.jpg
tt0317219	120000000	https://image.tmdb.org/t/p/original/qa6HCwP4Z15l3hpsASz3auugEW6.jpg

2. Title and runtime of movies Requete :

```
SELECT title, runtime FROM movie;
```

Retour :

title	runtime
The Son of Bigfoot	92
Finding Nemo	100
Shrek	90
Monsters University	104
The Super Mario Bros. Movie	92
Shrek the Third	93
L'apprenti Père Noël	80
Shazam!	132
Cars	117

3. 81 minutes movies Requete :

```
SELECT * FROM movie WHERE runtime = 81 ALLOW FILTERING;
```

Retour :

idimdb	budget	poster
tt0096842	0	https://image.tmdb.org/t/p/original/rcFi9JvJMjwxp4lK66v4KueqglW.jpg
tt0115491	0	https://image.tmdb.org/t/p/original/kTH7qEyvdDYMoQraQZm3LFKCge3.jpg
tt1216516	0	https://image.tmdb.org/t/p/original/hg1959yuBkHb4BKbIvETQSfxGCT.jpg
tt7367960	0	https://image.tmdb.org/t/p/original/dzN5xbp73TI7pjEqZ66ECNxdWJl.jpg
tt0114709	30000000	https://image.tmdb.org/t/p/original/uXDfjJbdP4ijW5hWSBrPr1Kpxab.jpg
tt0268380	59000000	https://image.tmdb.org/t/p/original/gLhHHZUzeseRXShoDyC4VqLgsNv.jpg

4. Movies which runtime is between 60 minutes and 80 minutes Re-
quete :

```
SELECT * FROM movie WHERE runtime >= 60 AND runtime <= 80 ALLOW FILTERING;
```

Retour :

idimdb	budget	poster
tt1754455	0	https://image.tmdb.org/t/p/original/ot2u06E1TjVCXkcoAEPSE1hHuTi.jpg
tt10635042	0	https://image.tmdb.org/t/p/original/aHMhFF7mT9tWyqBqeXBicjoYNS.jpg
tt0371823	0	https://image.tmdb.org/t/p/original/23bv0wf0S9fw347Yc68yPpkmd8i.jpg
tt0243585	120000000	https://image.tmdb.org/t/p/original/hjfeMLWqJY44mqjJKZSa6jx4Y1j.jpg
tt0288441	0	https://image.tmdb.org/t/p/original/eIkXJggVmzN4MyPgJ2kDjOy9lmQ.jpg
tt0450982	0	https://image.tmdb.org/t/p/original/a0VPQHPLNCWWmiXwjD7aJVRaH78.jpg
tt0283426	20000000	https://image.tmdb.org/t/p/original/DhuskQc9lSjPiYKr9hy0AZ7uiC.jpg
tt5042436	0	https://image.tmdb.org/t/p/original/uxdvJHE5V844dTlK2EmqRQCTYDb.jpg
tt12767498	0	https://image.tmdb.org/t/p/original/AwkmMTKJBAatbeAVhTwhcU3Pyp4.jpg
tt0033563	812000	https://image.tmdb.org/t/p/original/hKDdlslMtsU9JixAv5HR9biXlp.jpg

5. Number of movies Requete :

```
SELECT count(*) FROM movie;
```

Retour :

```
count
-----
463
```

6. Number of movies per runtime Prérequis :

Je dois changer la primary key pour que la runtime soit une colonne

```
DROP TABLE IF EXISTS movie;
```

```
CREATE TABLE movie(
  idimdb text,
  title text,
  release text,
  runtime int,
```

```

    budget int,
    revenue int,
    poster text,
    rating float,
    PRIMARY KEY (runtime, idimdb)
);

```

Requete :

```
SELECT runtime, count(*) FROM movie GROUP BY runtime;
```

Retour :

runtime	count
114	3
110	2
91	13
128	1
117	6
120	2
140	4
129	2
132	1
105	11

7. Average budget Requete :

```
SELECT avg(budget) FROM movie;
```

Retour :

system.avg(budget)
57077875

8. Most expensive movie Requete :

```
SELECT idimdb, title, MAX(budget) FROM movie;
```

Retour :

idimdb	title	system.max(budget)
tt1133985	Green Lantern	460000000

9. Titles of movies ordered in decreasing order by runtime Prérequis

: Avoir une clé de clustering commun or dans la table il n'y a pas d'élément commun a tous les films donc c'est pas possible.

Requete :

Retour :

10. Title of three movies that are longer than 140 minutes Requete
:

```
SELECT title FROM movie WHERE runtime > 140 LIMIT 3 ALLOW FILTERING;
```

Retour :

title

```
-----  
Pirates of the Caribbean: Dead Man's Chest  
Pirates of the Caribbean: The Curse of the Black Pearl  
Aquaman
```

5.2 Models

1. Genders with a set of movies that fit within the gender

```
CREATE TABLE gender (  
    idgender text,  
    name text,  
    movies set<text>,  
    PRIMARY KEY (idgender)  
);
```

2. Movies with a set of actors

```
CREATE TABLE movie_with_actors (  
    idmovie text,  
    title text,  
    actors set<text>,  
    PRIMARY KEY (idmovie)  
);
```

3. Studios with a set of movies that have been produced

```
CREATE TABLE studio (  
    idstudio text,  
    name text,  
    movies set<text>,  
    PRIMARY KEY (idstudio)  
);
```

4. Movies with all the elements (actors, studios and genders)

```
CREATE TABLE movie (  
    idmovie text,  
    title text,
```

```
release text,  
runtime int,  
budget int,  
revenue int,  
poster text,  
rating float,  
actors set<text>,  
studios set<text>,  
genders set<text>,  
PRIMARY KEY (idmovie)  
);
```